Results in Engineering 24 (2024) 103392

Contents lists available at ScienceDirect

Results in Engineering

journal homepage: www.sciencedirect.com/journal/results-in-engineering

Integrated internet of things (IoT) solutions for early fire detection in smart agriculture

Abdennabi Morchid [a,*], Zahra Oughannou [b], Rachid El Alami [a], Hassan Qjidaa [a], Mohammed Ouazzani Jamil [c], Haris M. Khalid [d,e]

[a] Department of Physics, Laboratory of Informatics, Signals, Automation and Cognitivism (LISAC), Faculty of Sciences Dhar El Mahraz, Sidi Mohamed Ben Abdellah University, BP 1796 Fes-Atlas 30003 Fez, Morocco
[b] De Advanced Systems Engineering Laboratory, ENSA, Ibn Tofail University, Kenitra, Morocco
[c] Systems and Sustainable Environment Laboratory (SED), Faculty of Engineering Sciences (FSI) Private University of Fez (UPF), Fez, Morocco
[d] College of Engineering and Information Technology, University of Dubai, Academic City, 14143 Dubai, United Arab Emirates
[e] Department of Electrical and Electronic Engineering Science, University of Johannesburg, Aukland Park 2006, South Africa

ARTICLE INFO

Keywords:
Smart agriculture
Internet-of-things (IoT)
Fire detection
Energy optimization
Sensors
Raspberry Pi
ThingSpeak
Data visualization
food safety
Sustainability

ABSTRACT

The integration of internet-of-things (IoT) technologies is proving crucial for optimizing crop management and monitoring in smart farming. The challenges posed by climate change which involve 1) increased pressure on natural resources, and 2) heightened food safety requirements, eventually call for innovative solutions to protect agricultural resources. Early fire detection, in particular, is essential to prevent major damage and ensure the sustainability of agricultural practices. This article presents an advanced IoT architecture for early fire detection. The proposed architecture integrates 1) smoke and flame sensors, 2) a Raspberry Pi for local processing, and 3) the ThingSpeak platform for data storage and visualization. The system is based on a three-layered architecture: 1) the IoT device layer, 2) the cloud layer, and 3) the application layer, enabling real-time collection and analysis of environmental data. Sensors collect real-time environmental data, which is then transmitted to ThingSpeak for storage. The ThingSpeak platform, combined with MATLAB visualization tools, enables continuous monitoring and in-depth trend analysis of historical data. The results show a clear improvement in accuracy and responsiveness in fire detection while contributing to the safety of agricultural resources and more sustainable management. The blend of IoT technologies, real-time data processing, and cloud-based visualization in this proposed detection system represents an important step towards safer and more resilient agriculture in the face of environmental risks.

1. Introduction

The rapid evolution of information and communication technologies has profoundly transformed various sectors, including agriculture. In a global context marked by climate change, increasing pressure on natural resources, and heightened food security requirements, agricultural practices must evolve to become more efficient and sustainable [1–5]. Advanced technologies such as the Internet of Things (IoT), artificial intelligence (AI), and cloud platforms offer promising solutions for improving the management and monitoring of agricultural resources, enabling a more proactive and informed approach [6–8]. In the agricultural sector, fire risk management represents a major challenge, potentially leading to significant resource losses and affecting food safety. Traditional fire detection methods may be inadequate for the speed and accuracy required for effective intervention. It is therefore essential to develop detection systems that enable real-time monitoring and more effective risk management, in order to protect crops and guarantee food safety [9,10]. This article focuses on the presentation and evaluation of an innovative IoT-based fire detection system aimed at improving risk management in agricultural environments. By integrating smoke and flame sensors with data processing performed by a Raspberry Pi, and using the ThingSpeak cloud platform for storage and visualization, the proposed system aims to offer a complete solution for proactive fire monitoring. The aim is to demonstrate how this technological integration can enhance the accuracy and responsiveness of fire detection while offering advanced tools for data analysis.

* Corresponding authors.
E-mail address: Abdennabi.morchid@usmba.ac.ma (A. Morchid).

https://doi.org/10.1016/j.rineng.2024.103392
Received 19 July 2024; Received in revised form 21 September 2024; Accepted 9 November 2024
Available online 10 November 2024
2590-1230/© 2024 The Author(s). Published by Elsevier B.V. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

The IoT plays a central role in the move toward intelligent agriculture by facilitating the connection and exchange of data between various devices and sensors [11,12]. In the system presented, smoke and flame sensors collect environmental data in real-time, while a Raspberry Pi provides local processing of this data. The ThingSpeak cloud platform is used for data storage and visualization, enabling continuous monitoring and in-depth trend analysis. The integration of these technologies enables more precise and proactive risk management, optimizing response to emergency situations. Smoke and flame sensors are at the heart of the detection system, providing real-time measurements of smoke concentrations and flame levels [13,14]. These data are essential for detecting signs of fire early and assessing their severity. The use of specialized sensors improves detection accuracy and ensures a rapid response, minimizing the risk of crop damage. The ThingSpeak cloud platform plays a crucial role in managing and analyzing the data collected. By storing data in the cloud, the system enables real-time visualization and historical trend analysis [15]. This ability to access and analyze data from any location facilitates proactive management and effective response to fire risks. Although this study does not specifically address AI, the integration of AI into future systems could enrich the detection system's ability to analyze collected data and provide even more accurate and useful insights [16].

This paper [17], greenhouses controlled by the IoT can offer considerable economic attractiveness. What's more, integrating photovoltaic panels in greenhouses located in hot climates can significantly reduce the need for additional energy for cooling. Further, this article [18] presents a unification of IoT and cloud, improving the sustainability and reliability of smart grids. It also proposes an IoT plan to reduce solar panel soiling, lower cleaning costs, and increase sustainable energy production. The study introduces an IoT-enabled solar sensor system, extending device life, saving energy, and providing valuable operational information. In addition, CloudIoT revolutionizes networks with predictive analytics, energy optimization, and intelligent design. Finally, the fog-based system suggests demand optimization, cost-effective pricing, and smart grid economics to promote sustainable energy. Additionally, this paper [19] presents an open IoT edge computing system for monitoring energy consumption in buildings. Implemented at the Faculty of Electromechanical Engineering of the University of Colima, this innovative platform optimizes the existing network infrastructure. Drawing on the emerging edge and fog computing paradigms, the solution focuses on the security of sensitive electrical data, offering detailed measurements of power, voltage, and current. Designed to be compatible with various IoT technologies, the study includes practical results that support and validate its effectiveness, highlighting the advantages of edge/fog computing over exclusively cloud-based architectures. The article also highlights the versatility of the platform for monitoring different variables, highlighting its applicability in a variety of contexts and monitoring needs. On the other hand, this article [20] presents the expansion of precision farming technologies from commercial agriculture to home gardening, highlighting the economic and health advantages as well as the physiological and psychological benefits for gardeners. It also addresses the problems of soil degradation and the lack of knowledge about soil properties and water chemistry content, leading to less efficient production of domestic plants. To remedy this, we have proposed a new connected sensor system, where smart watering can connect to a network of wireless sensors for soil and water analysis. Soil condition is measured using thick-film moisture and pH sensors. Further, this paper [21] presents an innovative approach to detecting sensor faults within wireless sensor networks (WSNs) used for smart irrigation. By combining an autoregressive model with a Kalman filter, this method overcomes the challenges posed by sensor failures that can alter the network's quality of service. The integration of these two tools makes it possible to capitalize on their respective strengths while taking into account the resource constraints of sensor nodes and the lack of ground truth information. The proposed technique stands out for its ease of implementation and low computational complexity while improving application reliability. Experiments and validations demonstrate the effectiveness of this combined approach for sensor fault detection in real WSN scenarios. In addition, this article [22] presents a real-time fire detection system based on the IoT, designed to protect forests. The project aims to develop a system capable of detecting fires and sending alerts to the authorities in a timely manner. Given the limited or non-existent bandwidth in forest areas, the system uses a GSM/GPRS module to communicate via a 2 G network. It integrates flame and temperature sensors to quickly identify fires, with machine learning algorithms to detect fire events over large areas. Data is transmitted to a cloud-based application, alerting authorities to abnormal temperature rises and predicting future fires.

This paper [23] presents a forest fire detection system based on smart sensors and IoT, focusing on machine learning methods. The proposed system integrates data from temperature sensors and smoke detectors to detect and categorize fires in real-time. The performance of the system is evaluated using a labeled dataset, explaining the advantages and disadvantages of the algorithms as well as potential difficulties for fire detection. The study also proposes future work to overcome these challenges and improve system performance. This research contributes to the development of reliable and effective innovations for forest fire identification, highlighting the need for further research in this crucial area. On the other hand, this article [24] presents an IoT-based fire detection system designed to detect and prevent damage caused by forest fires at an early stage. Due to the dry climate and winds in Korea, fires spread rapidly, making early detection and prevention essential. To meet this need, a real-time detection system was developed by combining a Raspberry Pi and a fisheye camera with a 220° lens. An image analysis algorithm was implemented using the OpenCV library to estimate the location of the fire by calculating the polar coordinates of the omnidirectional images. The continuous images acquired were transmitted to a Firebase database via the Raspberry Pi's Wi-Fi function, enabling the images to be analyzed to identify the fire's propagation path. Further, This paper [25] presents a monitoring and early warning system for forest fires, based on deep learning and the Internet of Things. Forest fire recognition has been improved by combining the size, color, and shape features of flames, smoke, and area. Complex features of fire images were extracted, optimizing input conversion with a fire risk prediction model based on an improved dynamic convolutional neural network. The proposed backpropagation neural network enhances system efficiency. This article [26] presents a real-time monitoring system with IoT and cloud, offering early fire detection, sustainable management of agricultural practices, enhanced food safety, and better protection of public health by reducing environmental risks. Furthermore, this article [27] presents the rise of mobile satellite systems as a key solution for Internet services in a connected world, with a focus on low-Earth orbit systems. Although promising for their wide coverage and robustness, low-orbit systems face challenges in terms of connectivity, stability, and mobility, making communication sometimes unreliable. On the other hand, this article [28] presents the use of AI and big data to optimize water, improve infrastructure management with predictive maintenance and leak detection, and monitor water resources in real-time thanks to the integration of satellites, drones, and IoT. This article [29] presents climate change indicators for urban areas, evaluated via satellite images and mathematical methods. It explores the use of AI to evaluate these indicators and make future projections, and proposes moving from low-resolution to high-resolution models. Further, this article [30] presents sixth-generation communication networks for smart cities, healthcare, transport, and agriculture. It uses optimization algorithms for power allocation, analyzes the security of data transmissions, and maximizes system capacity by power-domain multiplexing. On the other hand, this article [31] explores how EU manufacturing companies can achieve carbon neutrality and digital transformation. It provides advice on how to ensure the sustainability of the sector, forecasts future technologies using analytical tools, and

emphasizes that the future of the industry will be built gradually through concrete action and solidarity between EU countries. Furthermore, this article [32] proposes an optimization methodology for irrigation distribution, reducing CO2 emissions and using a multi-criteria solution with high correlation in the Gamma distribution. Further, this article [33] explores the use of biodegradable insulating materials, such as rice hulls and coconut fiber, for solar thermal collectors. Existing research in the field of real-time fire detection has several significant limitations in the context of smart farming. Firstly, these studies often rely on fire detection control methods that are not integrated and lack responsiveness to dynamic agricultural environments. In addition, they fail to effectively exploit advanced technologies such as the integration of IoT, smoke and flame sensors with the Raspberry Pi, cloud computing like ThingSpeak, and the MQTT communication protocol, all of which are crucial for accurate and reliable real-time monitoring. Another major shortcoming is the lack of presentation of a three-tier architecture IoT, cloud computing, and application - for effective management. Furthermore, the analysis of environmental data on the ThingSpeak cloud platform is often neglected, preventing proactive management of agricultural parameters, as well as the use of email alerts for immediate communication. These shortcomings demonstrate the need to develop more comprehensive solutions, based on modern technologies and enhanced safety protocols, to improve the reliability and efficiency of fire detection systems in smart farming.

This paper is structured as follows: Section 2 describes the material used and the methodology adopted for the study. Section 3 presents the results and discussion. Finally, Section 4 provides the conclusion and future work.

2. Materials and methods

This section presents the fundamental elements that motivated this research, as well as the main contribution of this study, which aims to improve risk management in agriculture through the integration of advanced technologies. Faced with the growing challenges of climate change and the need to enhance food security, it is becoming essential to develop intelligent systems capable of monitoring environmental conditions in real-time and preventing risks, such as fires. The main contribution of this work lies in the design and implementation of an early fire detection system, based on a three-layer architecture. This architecture comprises the IoT device layer, the cloud layer, and the application layer. The IoT devices capture data in real-time, while the cloud layer enables processing and storage. The application layer, meanwhile, facilitates data access and visualization, offering proactive and effective risk management. In addition, an algorithm has been developed to optimize the fire detection process and improve the system's responsiveness. Finally, the context and implementation of the system are also detailed, to describe the actual conditions under which the tests were carried out, thus ensuring relevant validation of the proposed system.

The motivation behind this research stems from the need to improve fire detection and management methods in agricultural environments. By combining advanced technologies such as IoT, high-precision sensors, and cloud solutions, the paper aims to propose an innovative solution that addresses current risk management challenges. The aim is to demonstrate how these technologies can be integrated to offer a comprehensive and effective solution, thus contributing to food security, the protection of agricultural resources, and the sustainability of farming practices. The following are the main contributions of this paper:

- The article presents an innovative IoT-based solution for early fire detection in the agricultural context, using smoke and flame sensors combined with a Raspberry Pi and the ThingSpeak cloud platform.
- The system is designed around a three-layer architecture - the IoT device layer, the cloud layer, and the application layer - offering a structured approach to collecting, processing, and visualizing data efficiently.
- The use of ThingSpeak and MATLAB enables real-time visualization of smoke and flame data, facilitating dynamic analysis and proactive fire risk management.
- Results show a significant improvement in fire detection responsiveness and accuracy, enhancing the ability to protect agricultural resources and minimize potential damage.
- By optimizing fire risk management, the system supports food security and promotes sustainable agricultural practices, reducing environmental impacts and improving the resilience of farming systems.

Intelligent agriculture integrates IoT technologies to improve crop management and monitoring. Early fire detection is crucial to protect agricultural resources and prevent catastrophic damage. This paper describes the architecture of an IoT-based fire detection system, using smoke and flame sensors, a Raspberry Pi for data processing, and the ThingSpeak cloud platform for data storage and visualization. As shown in Fig. 1, the system is based on a three-layer architecture: the IoT device, cloud, and application layers.

1) IoT device layer: The IoT device layer comprises the hardware components essential for fire detection. A Raspberry Pi $B+3$ acts as the central processing unit, managing sensor data acquisition and transmission to the cloud. Smoke and flame sensors are connected to the Raspberry Pi to detect smoke levels and the presence of flames in the environment. The network connection enables the Raspberry Pi to transmit the collected data to the ThingSpeak cloud platform for storage and analysis.

2) Cloud layer (ThingSpeak): ThingSpeak, as a cloud service, plays a crucial role in data storage and analysis. Data from smoke and flame sensors is sent to the cloud via API calls. ThingSpeak stores this data in a structured way and performs basic analysis and pre-processing to facilitate visualization and further use.

3) Application layer (ThingSpeak visualization): The application layer focuses on data visualization using ThingSpeak tools. ThingSpeak's integrated visualization tools and MATLAB are used to create graphical representations of the data, such as 3D surface graphs and heat maps. These visualizations enable real-time monitoring and analysis of trends and patterns in historical data.

This architecture combines the robustness of IoT devices for real-time data collection, the power of the cloud for secure storage and advanced analysis, and the intuitiveness of visualization tools for proactive fire risk management in smart agriculture. It represents a significant step towards safer, more sustainable and technologically advanced agriculture. Further, ThingSpeak is an essential cloud platform for collecting, analyzing, and visualizing data from IoT devices as part of this research. By centralizing data from smoke and flame sensors connected to a Raspberry Pi, ThingSpeak enables secure storage and real-time

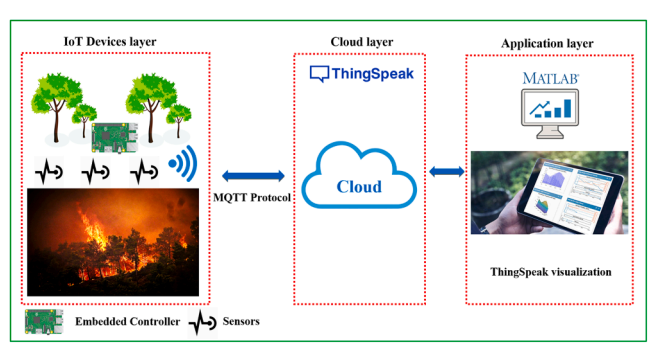

Fig. 1. Architecture of the proposed system.

accessibility. The platform offers advanced tools for data analysis, thanks in particular to its integration with MATLAB, enabling trends and anomalies to be detected quickly. Real-time data visualization via customizable graphics, such as 3D surface graphs, facilitates interpretation of environmental conditions and aids rapid decision-making. What's more, ThingSpeak can send notifications and alerts when predefined thresholds are exceeded, enhancing responsiveness and safety in environmental risk management. In this way, ThingSpeak plays a crucial role in improving proactive monitoring and risk management in smart agriculture, contributing to safer, more sustainable farming. In this context, the present article [33] describes the development of an intelligent and sustainable fire alarm system for smart cities. This system, based on IoT and advanced computing, addresses the limitations of existing systems by reducing installation costs and offering remote alerts. It consists of an ad hoc network of detection nodes, each equipped with an ESP8266-nodeMCU and various detect fire sensors. These nodes transmit their data to a central node, managed by a Raspberry Pi, via the Message Queuing Telemetry Transport (MQTT) protocol. In the event of fire detection, the system notifies the user and the fire department by SMS and call via a 4 G module. Users can also check the status of their home by SMS.

This pseudocode describes a fire detection system for smart farming as shown in Table 1, using a Raspberry Pi and smoke and flame sensors to monitor and report smoke and flame levels, while sending the data to the ThingSpeak cloud platform for visualization. First, the system defines two essential constants: the smoke threshold (SmokeThreshold) and the flame threshold (FlameThreshold). These values represent the limits beyond which an alert will be triggered to indicate a potential hazard. Next, several variables are initialized: smoke sensor data (smokeData) and flame sensor data (flameData), as well as the corresponding alert states (smokeAlert and flameAlert), initially set to false. The identifiers of the ThingSpeak channel (readChannelID) and the API key for reading the data (readAPIKey) are also specified. On the other hand, the Raspberry Pi and the smoke and flame sensors are then initialized to read the data. The system's main loop then begins. In this loop, data from the smoke and flame sensors are read out at regular intervals. If the smoke data exceeds the smoke threshold, a smoke alert is triggered by setting smokeAlert to true. Similarly, if flame data exceeds the flame threshold, a flame alert is triggered by setting flameAlert to true. Further, the data collected is then sent to ThingSpeak for storage and visualization. If the Raspberry Pi is connected to ThingSpeak, the data is displayed and sent to the platform. In the event of disconnection, the system attempts to reconnect. After reading and sending data, the system waits 2 s before resuming sensor reading. Further, MATLAB is configured for visualization to read smoke and flame data from ThingSpeak. The data is then used to create 3D surface graphs, making it easy to analyze and track trends in the data collected. This loop continues indefinitely, guaranteeing continuous, real-time smoke and flame levels monitoring in the farming environment. So, this algorithm ensures early detection of fires, enabling rapid intervention to protect crops and agricultural infrastructure. Data visualization via ThingSpeak and MATLAB provides a user-friendly interface for in-depth analysis and effective monitoring of critical parameters in the agricultural environment. The thresholds of the sensors used in our system, notably the smoke threshold (SmokeThreshold = 40) and the flame threshold (FlameThreshold = 50), were obtained through a series of experiments carried out under controlled conditions. For each type of sensor, several tests were carried out to define the optimum values to guarantee early detection while minimizing false alarms. The experiments involved exposing the sensors to different levels of smoke and flame, reproducing small-scale fire scenarios in an agricultural environment similar to that of the implementation site. The thresholds were adjusted progressively, according to the sensors' responses to real smoke and flame conditions. Each test was repeated several times to ensure the reliability of the measurements and the consistency of the results. In addition, comparisons with standard thresholds reported in the literature for similar sensors were made to validate our experimental choices. This approach enabled us to determine optimum values for rapid, accurate detection of fire hazards, thereby enhancing the system's effectiveness under real-life conditions. For the context and system implementation, the Fez region of Morocco (North Africa) is characterized by traditional agricultural practices typical of rural areas. Implementing our system in this context

Table 1
Algorithm: Pseudo-code of the proposed system.

```
// Define constants:
SmokeThreshold = 40
FlameThreshold = 50
// Variables:
smokeData ← Smoke sensor data
flameData ← Flame sensor data
smokeAlert = false ← Smoke alert status
flameAlert = false ← Flame alert status
readChannelID ← ThingSpeak channel ID
readAPIKey ← API key for reading data
// Initialize Raspberry Pi and sensors:
Initialize Raspberry Pi and smoke and flame sensors.
// Main loop:
While true:
// Read sensor data:
    smokeData = ReadSmokeSensor()
    flameData = ReadFlameSensor()
// Check thresholds and trigger alerts if necessary:
    smokeAlert = smokeData > SmokeThreshold
    flameAlert = flameData > FlameThreshold
// Send data to ThingSpeak:
    if (Raspberry Pi is connected to ThingSpeak):
        Display and send data to ThingSpeak
else:
        Try to reconnect
// Wait for next reading:
Wait (2 s)
// Visualize data on ThingSpeak with MATLAB:
    Configure MATLAB to read smoke and flame data from ThingSpeak
    Read data and create 3D surface plots
End while
```

enables us to assess its effectiveness under conventional farming conditions, often subject to challenges related to natural resource management and climatic hazards. Figs. 2 and 3 show the traditional farming methods used in this region. They highlight the specific site where our system has been deployed, providing a clear view of the experimental setting. This site, chosen for its representativeness of local farming practices, enabled the system to be tested under real-life conditions. These illustrations detail the exact location of the installation, providing an overview of the environmental and geographical variables to which the system had to adapt. This helps to contextualize the results obtained and to demonstrate the relevance of this approach in a traditional agricultural environment. Evaluating the system's effectiveness in this real-life setting highlights its potential for improving farm management in similar regions.

3. Results and discussion

This section presents the results obtained from tests carried out on the smoke and flame concentration measurement system. The data collected has been visualized using 3D surface plots to provide a detailed understanding of concentration variations over time. The graphs make it possible to analyze fluctuations in smoke and flame levels, offering insights into sensor behavior under real-time conditions. After running MATLAB code to visualize the smoke data, we produced a real-time 3D surface plot, shown in Fig. 4. This plot shows the measured smoke concentrations, ranging from 42 % to 43.3 %. The x-axis represents time, converted to digital format for greater accuracy, while the y-axis shows sample indices, providing a view of the measurement sequence. The z-axis shows smoke concentration levels. The 3D surface, colored according to smoke levels, is accompanied by a color scale to facilitate the interpretation of concentration intensities. The added grid enhances legibility and enables a more detailed analysis of temporal variations, providing a clear view of smoke concentration fluctuations and essential information for air quality assessment. For the flame data, a 3D surface plot was also generated, as shown in Fig. 5. This real-time plot illustrates flame concentration levels over the observation period, with a range of values from 15 % to 20 %. The x-axis represents time, converted to digital format to ensure temporal accuracy, while the y-axis corresponds to sample indices, providing a perspective on the sequence of measurements. The z-axis indicates flame concentration levels. The graph uses a colored surface to represent variations in flame concentration, accompanied by a color scale to indicate the intensity of the values. This visualization makes it possible to track fluctuations in flame levels with precision, providing a detailed view of trends and variations in flame levels over time, and is a valuable tool for analyzing flame behavior under various conditions. The 3D surface plots provided for smoke and flame data enable in-depth analysis of sensor behavior under real-life conditions. For smoke, the observed concentration range and variations over time reveal important trends that can inform air quality

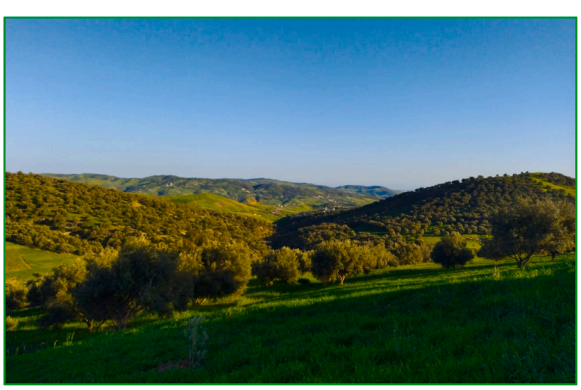

Fig. 2. Context and system implementation.

Fig. 3. Context and system implementation.

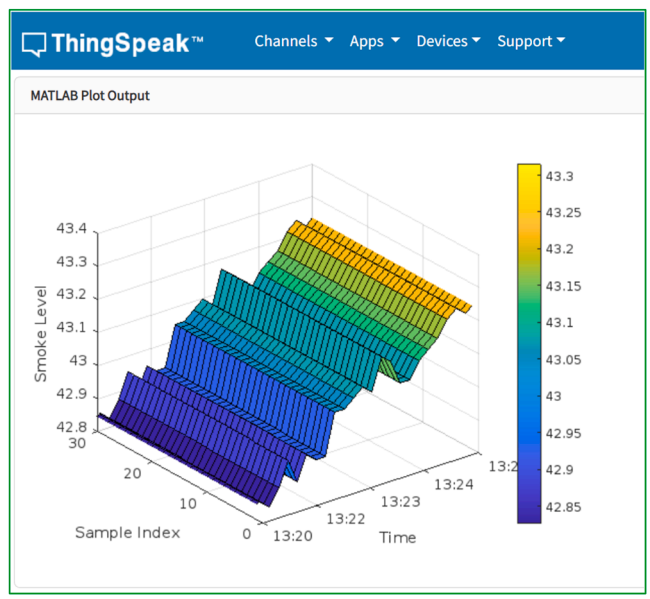

Fig. 4. Real-time 3D surface plot of smoke level.

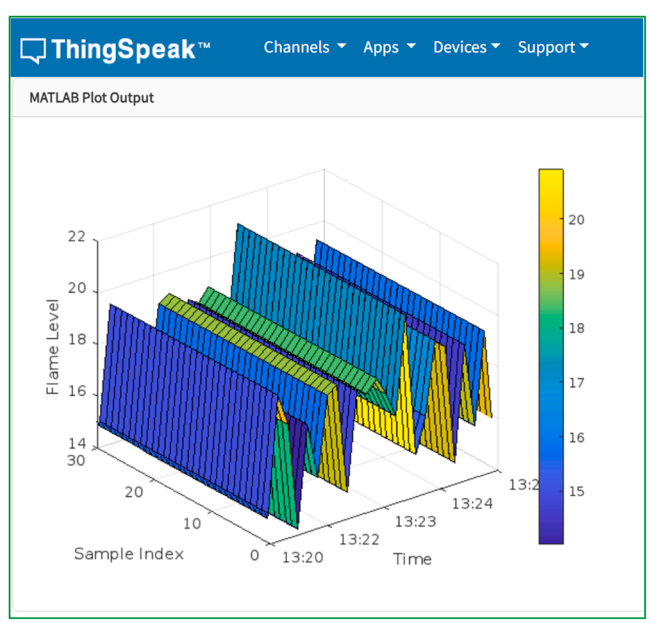

Fig. 5. Real-time 3D surface plot of flame level.

management strategies. Similarly, for flame, fluctuations in concentration levels offer insights into flame behavior and its interaction with the environment. These visualizations provide crucial insights for sensor performance evaluation and can guide future system optimizations. Data storage management and some basic analysis on ThingSpeak, it's crucial to specify that the platform performs various processing operations on the raw data collected by smoke and flame sensors, integrated into an on-board controller (Raspberry Pi). The data collected by this controller, using a specifically developed and implemented algorithm, is then transmitted to ThingSpeak via the MQTT protocol. On the platform, they undergo pre-processing processes such as normalization, anomaly filtering, and outlier correction. In addition, ThingSpeak enables basic calculations to be performed, such as rolling averaging, data aggregation, or the identification of peaks over defined periods. These pre-processing steps are essential to ensure accurate and reliable visualizations, facilitating real-time analysis of environmental data. This process optimizes the system's reactivity and enables more effective fire risk management as part of intelligent agriculture.

To clarify, real-time data management in the visualization process relies on regular and frequent updating of the information transmitted by the sensors. In the system shown here, the smoke and flame sensors send data at defined intervals (e.g. every 2 s) via the MQTT protocol to the embedded controller, which then transmits it to the ThingSpeak platform. This frequency ensures continuous, reactive monitoring. ThingSpeak, meanwhile, is configured to process and display this data in real-time. Thanks to its live analysis capabilities and calculations such as rolling averages or peak detection, the platform ensures accurate, up-to-date visualization with each new data transmission. This means greater reactivity in fire detection. In addition, mechanisms for checking anomalies and correcting outliers are integrated, ensuring that only reliable data is displayed, guaranteeing accurate and relevant visualization for rapid decision-making in the context of intelligent agriculture.

IoT plays a key role in the development of sustainable environmental remediation projects, enabling proactive resource management and rapid response to risks. It facilitates real-time monitoring of environmental conditions, early detection of risks, and optimization of agricultural practices. The data collected enables predictive analysis and better planning of interventions. By integrating these technologies, we can improve the sustainability of agricultural practices, minimize environmental impact, and support more effective remediation strategies. In the context of this study [34], this article explores the encapsulation of arsenic waste in building materials as a sustainable solution to avoid recontamination. On the other hand, real-time monitoring systems using IoT and cloud platforms, such as ThingSpeak, play a crucial role in public health and ecosystem management. They enable early detection of fires, helping to prevent major damage and smoke-related health risks. By optimizing farming practices, they promote sustainability and reduce environmental impact. What's more, they guarantee high standards of food production, ensuring food safety. Advanced analysis tools facilitate rapid, informed decision-making in the face of environmental conditions, helping to protect public health by reducing problems linked to pollution and climatic events. In this context this study [35], this paper presents a real-time monitoring system using IoT and cloud platforms, offering early detection of fires, optimization of agricultural practices, and enhanced food safety. It also facilitates rapid decision-making and improves public health protection by reducing the risks associated with pollution and climatic events.

Article [17] highlighted the economic appeal of IoT-controlled greenhouses and the positive impact of integrating photovoltaic panels in hot climates, reducing energy requirements for cooling. Furthermore, article [18] explores the unification of IoT and cloud, highlighting the increased sustainability and reliability of smart grids and proposes a plan to reduce solar panel fouling, improve sustainable energy production, and optimize grids through predictive analytics and intelligent design. Further, the paper [19] presents an open IoT edge computing system for monitoring energy consumption in buildings, optimizing existing network infrastructures with edge computing and fog computing to improve the security of sensitive electrical data. This platform demonstrates versatility in monitoring different variables, validated by practical results. On the other hand, article [20] discusses the expansion of precision agriculture technologies from commercial farming to home gardening, proposing a connected sensor system for intelligent soil and water analysis to solve problems related to soil degradation and inefficient watering. Furthermore, the article [21] proposed an innovative method for detecting sensor faults in wireless networks by combining an autoregressive model and a Kalman filter, improving network reliability despite sensor limitations and resource constraints. In addition, paper [22] presented an IoT-based real-time fire detection system, using a GSM/GPRS module to communicate in forested areas with limited bandwidth, integrating flame and temperature sensors and machine learning algorithms for early fire detection and warning. Further, article [23] highlights a forest fire detection system using intelligent sensors and machine learning methods to detect and categorize fires in real-time, with suggestions for improving system performance. Also, the article [24] described an IoT system for early detection of forest fires in a dry, windy climate, highlighting the importance of rapid response. Article [25] proposed a fire monitoring system based on deep learning and IoT, combining complex flame and smoke characteristics with a fire risk prediction model based on an enhanced dynamic convolutional neural network to optimize system efficiency.

In constat, this paper, which focuses on the visualization of smoke and flame concentration data using 3D surface graphics, has major implications in several key areas, such as agriculture, food safety, and environmental sustainability. The ability to monitor smoke and flame concentrations in real time plays a crucial role in environmental risk management. By integrating embedded systems such as the Raspberry Pi 3 $B+$, smoke and flame sensors, as well as IoT technologies, MQTT protocol, and the ThingSpeak cloud platform for data visualization, this study offers an innovative and comprehensive solution for proactive fire risk management. The impact, benefits, and importance of the research are presented below:

1) Impact on agriculture: In the agricultural sector, real-time monitoring of smoke and flame concentrations enables early detection of periods of air pollution caused by agricultural fires or burning. This is essential for protecting crops and optimizing management strategies. By reducing potential damage to crops, this approach helps to maintain high agricultural yields and ensure plant health. The precision of the data collected thanks to sensors and real-time processing via the Raspberry Pi 3 $B+$ enables rapid and effective intervention, minimizing agricultural losses.

2) Food safety: When it comes to food safety, the accuracy of smoke concentration data is decisive in assessing and mitigating the risks of contamination of foodstuffs by atmospheric pollutants. Efficient, real-time monitoring, made possible by the use of the ThingSpeak platform for data storage and visualization, helps maintain optimal growing conditions and guarantees high standards of food production. This strengthens the reliability of the food supply chain and ensures consumer safety.

3) Environmental sustainability: In terms of sustainability, this research provides essential tools for more proactive environmental management. By providing real-time data on pollution levels, the system enables a more accurate assessment of the environmental impacts associated with activities such as agricultural burning or forest fires. This facilitates rapid intervention and more effective risk management, supporting more sustainable practices and minimizing negative impacts on the ecosystem. Embedded systems like the Raspberry Pi 3 $B+$ play a key role in processing and transmitting data, ensuring continuous, reliable monitoring.

***4) Global significance*:** In conclusion, this research goes far beyond improving methods for monitoring smoke and flame concentrations.

It makes a significant contribution to food safety, agricultural risk management, and environmental sustainability. The use of advanced technologies, such as embedded systems, sensors, IoT, and cloud platforms, offers tangible benefits for the protection of natural resources and the resilience of food systems. These technological innovations enable more precise and proactive risk management, ensuring safer and more sustainable agriculture in the face of growing environmental challenges.

The results of this study enhance the relevance and potential impact on a global scale by addressing major global challenges such as climate change, sustainable agriculture, and environmental monitoring. By providing an advanced solution for early fire detection through the integration of technologies such as the IoT, smoke and flame sensors, Raspberry Pi, cloud platforms, and MQTT protocol, this research contributes directly to proactive environmental risk management. Fires, exacerbated by climate change, represent a serious threat to agricultural resources and food security. This study enables real-time monitoring, essential for mitigating the negative impacts of these events, by enhancing the responsiveness and accuracy of interventions. By facilitating more effective risk management, it supports global objectives linked to climate change adaptation and the protection of natural resources. In addition, the proposed methods are in line with international policies aimed at promoting sustainable agriculture. They offer tools for better management of agricultural resources, contributing to more resilient and environmentally friendly farming practices. By integrating innovative approaches to fire detection and management, this research also supports environmental monitoring efforts, enhancing the ability of agricultural systems to cope with current and future climate challenges. On the other hand, this study has some important limitations. Firstly, the lack of adequate training for farmers can pose a major problem, as data management via cloud computing requires a thorough understanding of technological systems. To overcome this difficulty, specific training programs for farmers are recommended, to familiarize them with the necessary tools and practices. Secondly, the system requires a stable Internet connection for real-time data transmission to the cloud. To solve this problem, it would be useful to develop alternative solutions such as enhanced Internet access points or temporary local storage devices that can synchronize data when the connection is re-established. Finally, the need for regular maintenance is crucial, especially when it comes to sensors and the devices' power source. To guarantee optimum performance, it is advisable to implement a preventive maintenance plan, including periodic checks and remote monitoring systems to proactively detect and resolve problems.

4. Conclusion and futur work

This research highlights the crucial importance of integrating IoT technologies for agricultural crop management and monitoring, particularly in the context of early fire detection. Using a combination of smoke and flame sensors, a Raspberry Pi 3 $B+$ for data processing, and the ThingSpeak cloud platform for real-time data storage and visualization, the proposed system demonstrated significant advantages in terms of responsiveness, accusracy, and efficiency. The results obtained show that the system enables real-time monitoring of smoke and flame concentrations, which is essential for protecting agricultural resources from fire. The ability to detect signs of fire early enables immediate intervention, reducing potential damage to crops and optimizing management strategies. This proactive approach helps to maintain high crop yields and ensure plant health, which is fundamental to ensuring food safety. This research goes far beyond improving methods for monitoring smoke and flame concentrations. It makes a significant contribution to food safety, agricultural risk management, and environmental sustainability. The use of advanced technologies, such as embedded systems, sensors, IoT, and cloud platforms, offers tangible benefits for the protection of natural resources and the resilience of food systems. These technological innovations enable more precise and proactive risk management, ensuring safer and more sustainable agriculture in the face of growing environmental challenges.

In future work, we plan to use images to integrate new technologies such as embedded systems, the IoT, and cloud computing for fire detection in smart agriculture. In addition, we will strive to exploit artificial intelligence (AI) to improve system accuracy and responsiveness. An important focus of future research will be the expansion of the system to monitor other environmental factors such as humidity and temperature. By integrating these parameters, we will be able to assess their impact on fire risk and crop health, offering a more comprehensive and integrated solution for smart agriculture. This approach will enable the development of more sophisticated management tools, contributing to better protection of agricultural resources and proactive risk management. These improvements aim to create a more global solution for sustainable agriculture, capable of responding to the complex challenges posed by contemporary agricultural environments. In addition, the authors of this article are keen to explore the possibility of collaborations with other disciplines or industries to further enrich the system. Working with agricultural experts to refine the system's parameters, or collaborating with policy-makers to ensure that the technology meets regulatory standards, could add significant value. Such collaborations would ensure that the system is not only technically advanced but also adapted to the real needs of the sector and compliant with current regulatory requirements.

CRediT authorship contribution statement

Abdennabi Morchid: Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Zahra Oughannou:** Writing – review & editing, Visualization, Validation, Software, Project administration, Funding acquisition, Formal analysis, Data curation. **Rachid El Alami:** Writing – review & editing, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Hassan Qjidaa:** Writing – review & editing, Visualization, Validation, Software, Project administration, Funding acquisition, Formal analysis, Data curation. **Mohammed Ouazzani Jamil:** Writing – review & editing, Visualization, Validation, Software, Funding acquisition, Formal analysis, Data curation. **Haris M. Khalid:** Writing – review & editing, Visualization, Validation, Software, Project administration, Funding acquisition, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

[1] S.K. Balasundram, R.R. Shamshiri, S. Sridhara, N. Rizan, The role of digital agriculture in mitigating climate change and ensuring food security: an overview, Sustainability. 15 (6) (Jan. 2023), https://doi.org/10.3390/su15065325.
[2] M. Javaid, A. Haleem, R.P. Singh, R. Suman, Enhancing smart farming through the applications of Agriculture 4.0 technologies, Internat. J. Intellig. Networks 3 (Jan. 2022) 150–164, https://doi.org/10.1016/j.ijin.2022.09.004.
[3] A. Blazquez-Soriano, R. Ramos-Sandoval, Information transfer as a tool to improve the resilience of farmers against the effects of climate change: the case of the Peruvian National agrarian innovation system, Agric. Syst. 200 (Jun. 2022) 103431, https://doi.org/10.1016/j.agsy.2022.103431.

[4] C.G. Bocean, A cross-sectional analysis of the relationship between digital technology use and agricultural productivity in EU countries, Agriculture 14 (4) (Apr. 2024), https://doi.org/10.3390/agriculture14040519.
[5] D. Kalfas, S. Kalogiannidis, O. Papaevangelou, K. Melfou, F. Chatzitheodoridis, Integration of technology in agricultural practices towards agricultural sustainability: a case study of Greece, Sustainability. 16 (7) (Jan. 2024), https://doi.org/10.3390/su16072664.
[6] B. Et-taibi, et al., Enhancing water management in smart agriculture: a cloud and IoT-Based smart irrigation system, Results. Eng. 22 (Jun. 2024) 102283, https://doi.org/10.1016/j.rineng.2024.102283.
[7] H. Shahab, M. Iqbal, A. Sohaib, F. Ullah Khan, M. Waqas, IoT-based agriculture management techniques for sustainable farming: a comprehensive review, Comput. Electron. Agric. 220 (May 2024) 108851, https://doi.org/10.1016/j.compag.2024.108851.
[8] A.A. Mana, A. Allouhi, A. Hamrani, S. Rehman, I. el Jamaoui, K. Jayachandran, Sustainable AI-based production agriculture: exploring AI applications and implications in agricultural practices, Smart Agricult. Techn. 7 (Mar. 2024) 100416, https://doi.org/10.1016/j.atech.2024.100416.
[9] S. Giannakidou, et al., Leveraging the power of internet of things and artificial intelligence in forest fire prevention, detection, and restoration: a comprehensive survey, Internet of Things 26 (Jul. 2024) 101171, https://doi.org/10.1016/j.iot.2024.101171.
[10] A. Abdusalomov, N. Baratov, A. Kutlimuratov, T.K. Whangbo, An Improvement of the Fire Detection and Classification Method Using YOLOv3 for Surveillance Systems, Sensors 21 (19) (Jan. 2021), https://doi.org/10.3390/s21196519.
[11] A. Morchid, R. El Alami, A.A. Raezah, Y. Sabbar, Applications of internet of things (IoT) and sensors technology to increase food security and agricultural Sustainability: benefits and challenges, Ain Shams Eng. J. 15 (3) (Mar. 2024) 102509, https://doi.org/10.1016/j.asej.2023.102509.
[12] A. Morchid, R. Jebabra, R.E. Alami, M. Charqi, B. Boukili, Smart agriculture for sustainability: the implementation of smart irrigation using real-time embedded system technology, in: 2024 4th International conference on innovative research in applied science, engineering and technology (IRASET), May 2024, pp. 1–6, https://doi.org/10.1109/IRASET60544.2024.10548972.
[13] A. Morchid et al., 'Fire safety system implementation for controlling and monitoring a siren in smart farm using gas sensor and flame sensor', in Digital Technologies and Applications, S. Motahhir and B. Bossoufi, Eds., Cham: Springer International Publishing, 2022, pp. 733–742. doi: 10.1007/978-3-031-01942-5_73.
[14] A. Morchid, M. El Alaoui, R. El Alami, H. Qjidaa, K. El Khadiri, and Y. Mehdaoui, 'Design and realization of fire safety system for controlling and monitoring a siren using arduino Uno', in Digital technologies and applications, S. Motahhir and B. Bossoufi, Eds., Cham: Springer International Publishing, 2021, pp. 433–445. doi: 10.1007/978-3-030-73882-2_40.
[15] A. Morchid, I.G. Muhammad Alblushi, H.M. Khalid, R. El Alami, S.R. Sitaramanan, S.M. Muyeen, High-technology agriculture system to enhance food security: a concept of smart irrigation system using Internet of Things and cloud computing, J. Saudi Soc. Agricult. Sci. (Feb. 2024), https://doi.org/10.1016/j.jssas.2024.02.001.
[16] A. Morchid, M. Marhoun, R. El Alami, B. Boukili, Intelligent detection for sustainable agriculture: a review of IoT-based embedded systems, cloud platforms, DL, and ML for plant disease detection, Multimed. Tools. Appl. (Feb. 2024), https://doi.org/10.1007/s11042-024-18392-9.
[17] S. Al-Naemi, A. Al-Otoom, Smart sustainable greenhouses utilizing microcontroller and IOT in the GCC countries; energy requirements & economical analyses study for a concept model in the state of Qatar, Results. Eng. 17 (Mar. 2023) 100889, https://doi.org/10.1016/j.rineng.2023.100889.
[18] A. Rajagopalan, et al., Empowering power distribution: unleashing the synergy of IoT and cloud computing for sustainable and efficient energy systems, Results. Eng. 21 (Mar. 2024) 101949, https://doi.org/10.1016/j.rineng.2024.101949.
[19] D.A. Verde Romero, E. Villalvazo Laureano, R.O. Jiménez Betancourt, E. Navarro Álvarez, An open source IoT edge-computing system for monitoring energy consumption in buildings, Results. Eng. 21 (Mar. 2024) 101875, https://doi.org/10.1016/j.rineng.2024.101875.
[20] A. Rabak, et al., Sensor system for precision agriculture smart watering can, Results. Eng. 19 (Sep. 2023) 101297, https://doi.org/10.1016/j.rineng.2023.101297.
[21] N. Jihani, M.N. Kabbaj, M. Benbrahim, Kalman filter based sensor fault detection in wireless sensor network for smart irrigation, Results. Eng. 20 (Dec. 2023) 101395, https://doi.org/10.1016/j.rineng.2023.101395.
[22] V. Venkataramanan, G. Kavitha, M.R. Joel, J. Lenin, Forest Fire Detection and Temperature Monitoring Alert using IoT and Machine Learning Algorithm, in: 2023 5th International Conference on Smart Systems and Inventive Technology (ICSSIT), Jan. 2023, pp. 1150–1156, https://doi.org/10.1109/ICSSIT55814.2023.10061086.
[23] C. Nagolu, C. Cheekula, D. Sai Kiran Thota, K. Padmanaban, D. Bhattacharyya, Real-time forest fire detection using IoT and smart sensors, in: 2023 International conference on inventive computation technologies (ICICT), Apr. 2023, pp. 1441–1447, https://doi.org/10.1109/ICICT57646.2023.10134063.
[24] C.H. Lee, W.H. Lee, S.M. Kim, Development of iot-based real-time fire detection system using raspberry pi and fisheye camera, App. Sci. 13 (15) (Jan. 2023), https://doi.org/10.3390/app13158568.
[25] S. Zheng, et al., An accurate forest fire recognition method based on improved BPNN and IoT, Remote Sens. 15 (9) (Jan. 2023), https://doi.org/10.3390/rs15092365.
[26] S. Anbazhagan, R.K. Mugelan, Energy efficiency optimization of NB-IoT using integrated Proxy & ERAI technique, Results. Eng. 23 (2024) 102419, https://doi.org/10.1016/j.rineng.2024.102419.
[27] I. Shayea, A.A. El-Saleh, M. Ergen, B. Saoud, R. Hartani, D. Turan, A. Kabbani, Integration of 5G, 6G and IoT with Low Earth Orbit (LEO) networks: opportunity, challenges and future trends, Results. Eng. 23 (2024) 102409, https://doi.org/10.1016/j.rineng.2024.102409.
[28] H. Kamyab, T. Khademi, S. Chelliapan, M. SaberiKamarposhti, S. Rezania, M. Yusuf, M. Farajnezhad, M. Abbas, B. Hun Jeon, Y. Ahn, The latest innovative avenues for the utilization of artificial Intelligence and big data analytics in water resource management, Results. Eng. 20 (2023) 101566, https://doi.org/10.1016/j.rineng.2023.101566.
[29] B. Sirmacek, R. Vinuesa, Remote sensing and AI for building climate adaptation applications, Results. Eng. 15 (2022) 100524, https://doi.org/10.1016/j.rineng.2022.100524.
[30] A. N, K s P, Meta-algorithmic optimized power allocation in cybertwin-based sixth generation cooperative communication system, Results. Eng. 23 (2024) 102740, https://doi.org/10.1016/j.rineng.2024.102740.
[31] A.K. Mallik, The future of the technology-based manufacturing in the European Union, Results. Eng. 19 (2023) 101356, https://doi.org/10.1016/j.rineng.2023.101356.
[32] M.A. Garcia-Espinal, F.J. Sanchez-Romero, M. Perez-Sanchez, P.A. Lopez-Jimenez, Enhancing sustainability in irrigation networks: a multicriteria method for optimizing flow distribution and reducing environmental impact, Results. Eng. 23 (2024) 102609, https://doi.org/10.1016/j.rineng.2024.102609.
[33] A. Mahgoub, N. Tarrad, R. Elsherif, L. Ismail, A. Al-Ali, Fire alarm system for smart cities using edge computing, in: 2020 IEEE International conference on informatics, IoT, and enabling technologies (ICIoT), 2020, pp. 597–602, https://doi.org/10.1109/ICIoT48696.2020.9089653.
[34] S. Koley, Arsenic calamity in India's West Bengal: a critical review of mitigation scenarios, in: Proceedings of the institution of civil engineers - Engineering sustainability 176, 2023, pp. 198–213, https://doi.org/10.1680/jensu.21.00066.
[35] H. Luo, H. Cai, H. Yu, Y. Sun, Z. Bi, L. Jiang, A short-term energy prediction system based on edge computing for smart city, Fut. Generat.Comput. Syst. 101 (2019) 444–457, https://doi.org/10.1016/j.future.2019.06.030.